1. Grey Level Transformation

(a) Purpose: Maps input intensity values (r) to output intensity values (s) using a transformation function to alter contrast, dynamic range, or overall brightness.

(b) Computation: Transformation $s = 255 - r$ for input $r = [50, 100, 150, 200]$:

- $r = 50 \rightarrow s = 255 - 50 = 205$
- $r = 100 \rightarrow s = 255 - 100 = 155$
- $r = 150 \rightarrow s = 255 - 150 = 105$
- $r = 200 \rightarrow s = 255 - 200 = 55$

Transformed Values: **[205, 155, 105, 55]**

2. Log Transformation

(a) Explanation: Expands the range of low-intensity (dark) pixels while compressing higher-level intensity values, making details in dark regions much clearer.

(b) Computation: Given $s = c \cdot log_{10}(1 + r)$ where $c = 10$ for $r = [0, 10, 50, 100]$:

- $r = 0 \rightarrow s = 10 \cdot log_{10}(1) = 0$
- $r = 10 \rightarrow s = 10 \cdot log_{10}(11) \approx 10 \cdot 1.0414 = 10.41$
- $r = 50 \rightarrow s = 10 \cdot log_{10}(51) \approx 10 \cdot 1.7076 = 17.08$
- $r = 100 \rightarrow s = 10 \cdot log_{10}(101) \approx 10 \cdot 2.0043 = 20.04$

Transformed Values: **[0, 10.41, 17.08, 20.04]**

3. Histogram Processing

(a) Role: Equalization redistributes intensity levels uniformly across the full dynamic range to enhance global contrast.

(b) Computation (3-bit image, L-1 = 7, Total pixels N = 16):

- **CDF Calculation:**
  - $r=0$ (freq 2) → CDF = 2
  - $r=1$ (freq 2) → CDF = 2 + 2 = 4
  - $r=2$ (freq 4) → CDF = 4 + 4 = 8
  - $r=3$ (freq 8) → CDF = 8 + 8 = 16
- **Equalized Intensities $s_k = round((CDF / N) \times 7)$:**
  - $s_0 = round((2 / 16) \times 7) = round(0.875) = 1$
  - $s_1 = round((4 / 16) \times 7) = round(1.75) = 2$
  - $s_2 = round((8 / 16) \times 7) = round(3.5) = 4$
  - $s_3 = round((16 / 16) \times 7) = round(7.0) = 7$

Equalized Output Mapping: **0 → 1, 1 → 2, 2 → 4, 3 → 7**

4. Spatial Filtering (Smoothing)

(a) Purpose: Reduces high-frequency noise and sharp intensity transitions by calculating local neighborhood averages.

(b) Computation: 3×3 averaging filter on region [10, 20, 30; 20, 30, 40; 30, 40, 50]:

$$\text{Center Value} = (10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50) / 9 = 270 / 9 = 30$$

New Center Pixel: **30**

5. Spatial Filtering (Sharpening)

(a) Purpose: Highlights fine details and spatial discontinuities using second-order derivative operators (Laplacian).

(b) Computation: Apply Laplacian mask *[0 -1 0; -1 4 -1; 0 -1 0]* to matrix *[10 10 10; 10 20 10; 10 10 10]*:

$$\text{Output} = (0\times10) + (-1\times10) + (0\times10) + (-1\times10) + (4\times20) + (-1\times10) + (0\times10) + (-1\times10) + (0\times10)$$
$$= -10 - 10 + 80 - 10 - 10 = 40$$

Output at Center: **40**

6. Frequency Domain Filtering

(a) Differentiation: Low-pass filters retain low frequencies to blur images and remove high-frequency noise. High-pass filters attenuate low frequencies to sharpen edges.

(b) Computation: Given $F = [100, 80, 40, 20]$, retaining only the first two components sets remaining elements to zero:

Filtered Output: **[100, 80, 0, 0]**

7. Thresholding

(a) Global Thresholding: Partitions an image into foreground and background by comparing all pixel values against a single global threshold value T.

(b) Assignment: Given pixel value $r = 200$ and threshold $T = 100$:

Since $200 \geq 100$, pixel is assigned **1** .

8. Edge Detection

(a) Concept: Locates boundaries of objects by detecting significant local intensity variations or spatial gradients.

(b) Sobel Horizontal Kernel: Apply $[-1\ -2\ -1;\ 0\ 0\ 0;\ 1\ 2\ 1]$ to matrix $[10\ 10\ 10;\ 20\ 20\ 20;\ 30\ 30\ 30]$:

$$\text{Gradient} = (-1\times10) + (-2\times10) + (-1\times10) + (0\times20) + (0\times20) + (0\times20) + (1\times30) + (2\times30) + (1\times30)$$
$$= -10 - 20 - 10 + 0 + 0 + 0 + 30 + 60 + 30 = 80$$

Gradient at Center: **80**

9. Region-Based Segmentation

(a) Region Growing: Groups neighboring pixels into larger regions based on defined similarity criteria starting from seed points.

(b) Computation: Seed intensity = 50, threshold diff = 5 ⇒ Allowed range $[45, 55]$.

Given pixel intensities: $[1, 0, 2]$.

None of the pixels fall within $[45, 55]$.

Identified Region Pixels: **No pixels belong to the region**

10. Combined Enhancement & Segmentation

(a) Explanation: Smoothing reduces high-frequency noise that could otherwise cause spurious edge responses or false regional segments.

(b) Step-by-step:

- Averaging result = 48
- Threshold check ($T = 50$): Since $48 < 50$, output is set to 0.

Final Segmented Output: **0**

11. Power-Law (Gamma) Transformation

(a) Role: Controls non-linear brightness mapping to match physical display devices or highlight specific intensity bands.

(b) Justification: Setting $\gamma = 0.5$ (< 1) stretches dark intensity values, brightening shadowed details.

(c & d) Step-by-step Computation ($s = 1 \cdot r^{0.5} = \sqrt{r}$):

- $r = 25 \rightarrow s = \sqrt{25} = 5$
- $r = 100 \rightarrow s = \sqrt{100} = 10$
- $r = 225 \rightarrow s = \sqrt{225} = 15$

Transformed Values: **[5, 10, 15]**

(e) Interpretation: Expands lower-intensity pixel spacing, brightening dark regions.

12. Contrast Stretching

(a) Need: Expands narrow input intensity ranges across the entire dynamic range of the display.

(b) Justification: Linear scaling to $[0, 255]$ maximizes dynamic visual contrast.

(c & d) Computation ($s = [(r - 50) / (150 - 50)] \times 255$):

- $r = 50 \rightarrow s = (0 / 100) \times 255 = 0$
- $r = 100 \rightarrow s = (50 / 100) \times 255 = 127.5$
- $r = 150 \rightarrow s = (100 / 100) \times 255 = 255$

Transformed Values: **[0, 127.5, 255]**

(e) Interpretation: Significantly boosts overall image contrast, separating light and dark elements clearly.

13. Histogram Equalization (4-bit Image)

(a) Need: Maximizes global contrast when pixel intensities are clustered in a small sub-range.

(b) Justification: Cumulative distribution function (CDF) produces a monotonic mapping function.

(c & d) Computation (Total N = 26, Max Level L-1 = 15):

- $r = 0$ (freq 4) → CDF = 4 → $s_0 = round((4 / 26) \times 15) = round(2.307) = 2$
- $r = 1$ (freq 6) → CDF = 10 → $s_1 = round((10 / 26) \times 15) = round(5.769) = 6$
- $r = 2$ (freq 10) → CDF = 20 → $s_2 = round((20 / 26) \times 15) = round(11.538) = 12$
- $r = 3$ (freq 6) → CDF = 26 → $s_3 = round((26 / 26) \times 15) = round(15.0) = 15$

New Intensity Levels: **0 → 2, 1 → 6, 2 → 12, 3 → 15**

(e) Interpretation: Broadens intensity distribution from range [0,3] to [2,15], drastically boosting contrast.

14. Median Filtering

(a) Noise Characteristics: Salt-and-pepper noise presents extreme high/low intensity impulse spikes.

(b) Justification: Non-linear median filtering removes extreme outliers without blurring sharp edges.

(c & d) Neighborhood Sorting:

Neighborhood: *[10, 200, 10, 200, 50, 200, 10, 200, 10]*
Sorted: *[10, 10, 10, 10, **50**, 200, 200, 200, 200]*
Center Value: **50**

(e) Interpretation: Completely eliminates impulse noise while restoring original regional pixel value.

15. Gaussian Smoothing

(a) Objective: Reduces high-frequency noise using isotropic spatial weighted averaging.

(b) Justification: Provides smooth, continuous spatial falloff without boxy ringing artifacts.

(c & d) Computation: Mask weights sum = 16.

$$\text{Weighted Sum} = (1\times10 + 2\times20 + 1\times10 + 2\times20 + 4\times40 + 2\times20 + 1\times10 + 2\times20 + 1\times10) = 360$$
$$\text{Center Pixel} = 360 / 16 = 22.5$$

Center Pixel: **22.5**

(e) Interpretation: Smoothly suppresses spatial noise while heavily weighting local center context.

16. High-Pass Filtering

(a) Need: Amplifies fine spatial details and high-frequency edge transitions.

(b) Justification: High-pass masks attenuate low-frequency background levels, leaving edge contours.

(c & d) Matrix Application on Q15 matrix: Mask *[-1 -1 -1; -1 8 -1; -1 -1 -1]*:

$$\text{Weighted Sum} = (-1\times10 - 1\times20 - 1\times10 - 1\times20 + 8\times40 - 1\times20 - 1\times10 - 1\times20 - 1\times10) = 200$$

Center Value: **200**

(e) Interpretation: Significantly accentuates local intensity sharp changes.

17. Frequency Domain High-Pass

(a) High-Frequency Importance: Contains critical fine structural information and edge boundaries.

(b) Justification: Suppresses low-frequency illumination trends to highlight edges.

(c & d) Filtering $F = [10, 20, 80, 100]$: Zero out low frequencies (first two components).
Output: **[0, 0, 80, 100]**

(e) Interpretation: Isolates rapid intensity transitions while filtering out baseline lighting.

18. Adaptive Thresholding

(a) Challenge: Global thresholding fails under non-uniform spatial illumination.

(b) Justification: Evaluates thresholds locally based on neighboring region statistics.

(c & d) Binarization (Mean = 100 for *[60, 80, 120, 140]*):

- $60 < 100 \rightarrow 0$
- $80 < 100 \rightarrow 0$
- $120 \geq 100 \rightarrow 1$
- $140 \geq 100 \rightarrow 1$

Binary Output: [0, 0, 1, 1]

(e) Interpretation: Accurately segments pixels despite regional illumination differences.

19. Edge Detection (Vertical Sobel)

(a) Objective: Detects vertical edge boundaries along horizontal intensity gradients.

(b) Justification: Combines differentiation with spatial smoothing perpendicular to the edge direction.

(c & d) Apply Vertical Sobel *[-1 0 1; -2 0 2; -1 0 1]* to matrix *[10 20 30; 10 20 30; 10 20 30]*:

$$\text{Output} = (-1\times10 + 0 + 1\times30) + (-2\times10 + 0 + 2\times30) + (-1\times10 + 0 + 1\times30) = 20 + 40 + 20 = 80$$

Center Value: 80

(e) Interpretation: Confirms a strong vertical edge transition with horizontally increasing intensity.

20. Prewitt Edge Detection

(a) Purpose: Detects directional intensity gradients across pixel neighborhoods.

(b) Justification: Computationally simpler than Sobel as it avoids non-uniform spatial weighting.

(c & d) Apply Prewitt Kernel *[-1 0 1; -1 0 1; -1 0 1]* to Q19 Matrix:

$$\text{Output} = (-1\times10 + 1\times30) + (-1\times10 + 1\times30) + (-1\times10 + 1\times30) = 20 + 20 + 20 = 60$$

Output Value: 60

(e) Interpretation: Effectively computes linear directional gradient magnitude.

21. Region Splitting

(a) Problem: Non-homogeneous pixel sets require division into uniform sub-regions.

(b) Justification: Hierarchical splitting divides regions failing a uniformity test.

(c & d) Splitting at Threshold 80 for *[40, 45, 100, 105]*:
Region 1: [40, 45]
Region 2: [100, 105]

(e) Interpretation: Successfully separates two distinct homogenous clusters.

22. Region Merging

(a) Criteria: Combines adjacent sub-regions if their mean difference is below a similarity threshold.

(b) Justification: Recombines over-segmented sub-regions with near-identical properties.

(c & d) Evaluation for $R_1 = [45, 50]$ (Mean = 47.5) and $R_2 = [52, 55]$ (Mean = 53.5) with threshold 5:
Mean difference $= |53.5 - 47.5| = 6.0$. Since $6.0 > 5$, regions do not merge.
Merged Region: **[45, 50] and [52, 55] remain separate**

(e) Interpretation: Preserves distinct boundaries when regional variation exceeds threshold limits.

23. Edge Linking

(a) Problem: Noise and uneven contrast cause broken or fragmented edge segments.

(b) Justification: Connects adjacent edge pixels based on spatial proximity and orientation continuity.

(c & d) Linking on *[1, 0, 1, 0, 1]*:
Gaps filled between strong adjacent edges: **[1, 1, 1, 1, 1]**

(e) Interpretation: Establishes complete, uninterrupted boundary contours.

24. Low-Pass Filtering (Frequency Domain)

(a) Requirement: Attenuates rapid high-frequency pixel variations to smooth the overall image.

(b) Justification: Passes low-frequency background components while cutting high-frequency noise spikes.

(c & d) Filter Application to $F = [200, 150, 50, 20]$:
Cut high frequencies (> 100): **[200, 150, 0, 0]**

(e) Interpretation: Smooths image details and suppresses high-frequency noise.

25. Laplacian Sharpening (Variant)

(a) Need: Sharpens image details by adding second-derivative high-frequency details back to original intensities.

(b) Justification: Mask *[0 -1 0; -1 5 -1; 0 -1 0]* performs combined edge extraction and original image addition.

(c & d) Application to Matrix *[10 20 10; 20 40 20; 10 20 10]*:

$$\text{Center} = (-1\times20) + (-1\times20) + (5\times40) + (-1\times20) + (-1\times20) = -80 + 200 = 120$$

Center Output: **120**

(e) Interpretation: Significantly boosts central pixel contrast relative to surrounding pixels.

26. Histogram Matching

(a) Need: Adjusts the intensity distribution of an input image to match a specified reference histogram.

(b) Justification: Ensures tonal consistency across multiple images taken under different lighting.

(c & d) Mapping *[10, 50, 100]* (linear map to *[0, 255]*):

- $10 \rightarrow 0$
- $50 \rightarrow [(50 - 10) / 90] \times 255 = 113.33$
- $100 \rightarrow 255$

Mapped Values: **[0, 113.33, 255]**

(e) Interpretation: Standardizes overall dynamic range to match reference appearance.

27. Noise Reduction Comparison

(a) Noise Type: Impulse (Salt-and-Pepper) noise producing extreme pixel values.

(b) Comparison: Averaging spreads noise intensity into surrounding pixels; median completely removes extreme outliers.

(c & d) Filter application on *[10, 200, 10]*:

- **Averaging:** $(10 + 200 + 10) / 3 = 73.33$
- **Median:** Sorted $[10, 10, 200] \rightarrow 10$

Outputs: **Averaging: 73.33 | Median: 10**

(e) Interpretation: Median filter completely suppresses impulse noise without distorting background level.

28. Combined Filtering

(a) Need: Smoothing before sharpening prevents noise from being mistakenly amplified during edge enhancement.

(b) Justification: Low-pass median filtering removes noise spikes, allowing sharpening to focus purely on structural edges.

(c & d) Sequential Filtering on *[10, 200, 10]*:

- Step 1 (Median smoothing): Output = 10
- Step 2 (Sharpening): Operates on uniform 10 = 10

Final Output: **10**

(e) Interpretation: Effectively cleans noise while preserving valid structural boundaries.

29. Multi-Threshold Segmentation

(a) Interpretation: Segments complex scenes into multiple regions using multiple decision boundaries.

(b) Justification: Dual thresholds ($T_1 = 50, T_2 = 100$) classify pixels into 3 intensity regions.

(c & d) Application to ***[20, 60, 120, 180]*****:**

- $20 < 50 \rightarrow Class\ 0$
- $50 \le 60 \le 100 \rightarrow Class\ 1$
- $120 > 100 \rightarrow Class\ 2$
- $180 > 100 \rightarrow Class\ 2$

Class Assignments: **[0, 1, 2, 2]**

(e) Interpretation: Successfully groups image components into dark, mid-tone, and bright regions.

30. Boundary Detection

(a) Interpretation: Identifies and isolates outer structural perimeters of segmented target objects.

(b) Justification: Combines gradient magnitude detection with edge linking to form continuous closed boundaries.

(c & d) Pipeline Result:
Computes directional spatial gradients followed by neighbor linking to generate unbroken closed boundary contours.

(e) Interpretation: Delivers continuous boundary contours ready for object recognition and shape extraction.